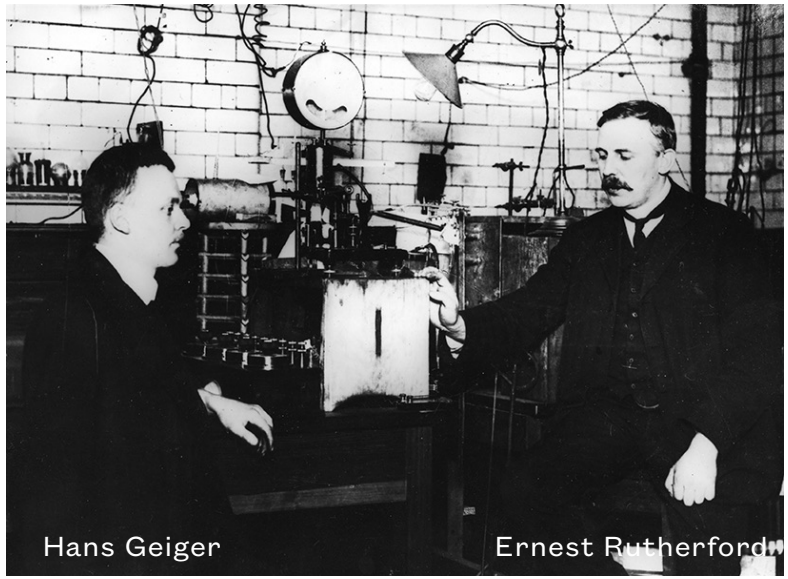
Modern Physics: Part 3

SUZIE SHEEHY, SUMMER SEMESTER

Objectives:

- How the nucleus was discovered
- Why in classical electromagnetism the nucleus of the atom cannot be a stable system
- Understand an electronVolt eV
- Explain wave particle duality



Sir Ernest Rutherford:

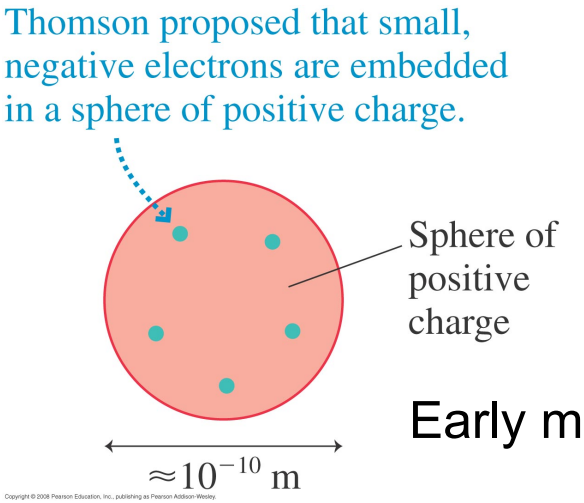
The New Zealand farmers son who went on to be a Nobel prize winner, President of the Royal Society and ‘father of nuclear physics’

Rutherford and the discovery of the nucleus §42.1

The discovery of the electrons emanating from atoms suggested that atoms were composed of charged particles. However, atoms were electrically neutral overall

→ # +ve charges = # -ve charges

JJ Thomson's Plum Pudding model: (Thomson was Rutherfords PhD supervisor at Cambridge)



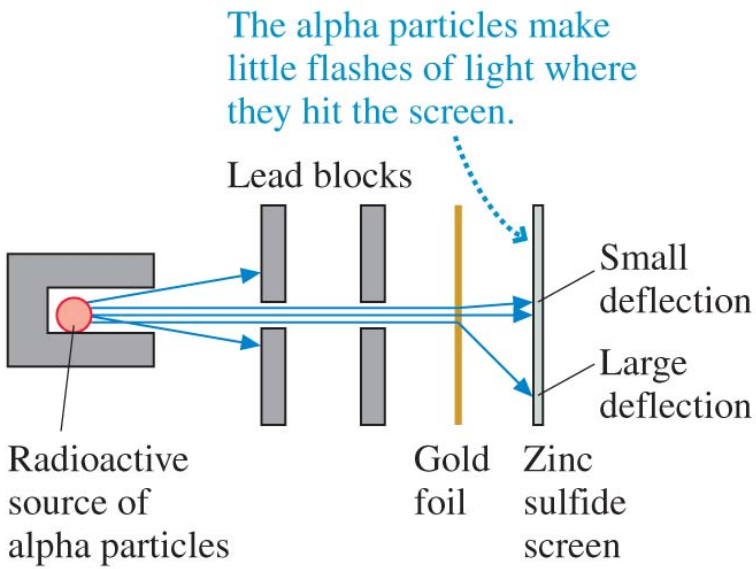
Early model of the atom.

Rutherford, Marsden, Geiger: the discovery of the nucleus

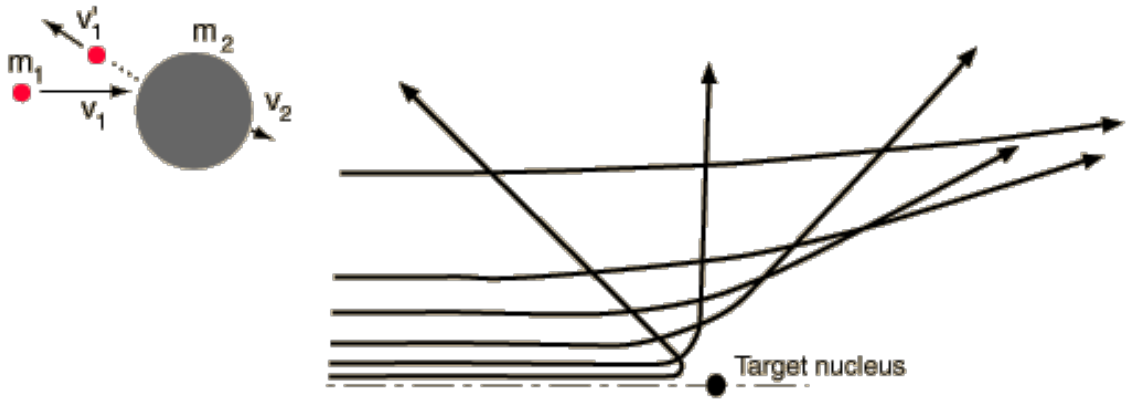
- In 1909 Hans Geiger and Ernest Marsden experiment: alpha particles deflected from gold foil at *large* angles.

- alpha particles deflected at large angles, but a very few ***were reflected almost straight backward toward the source!***

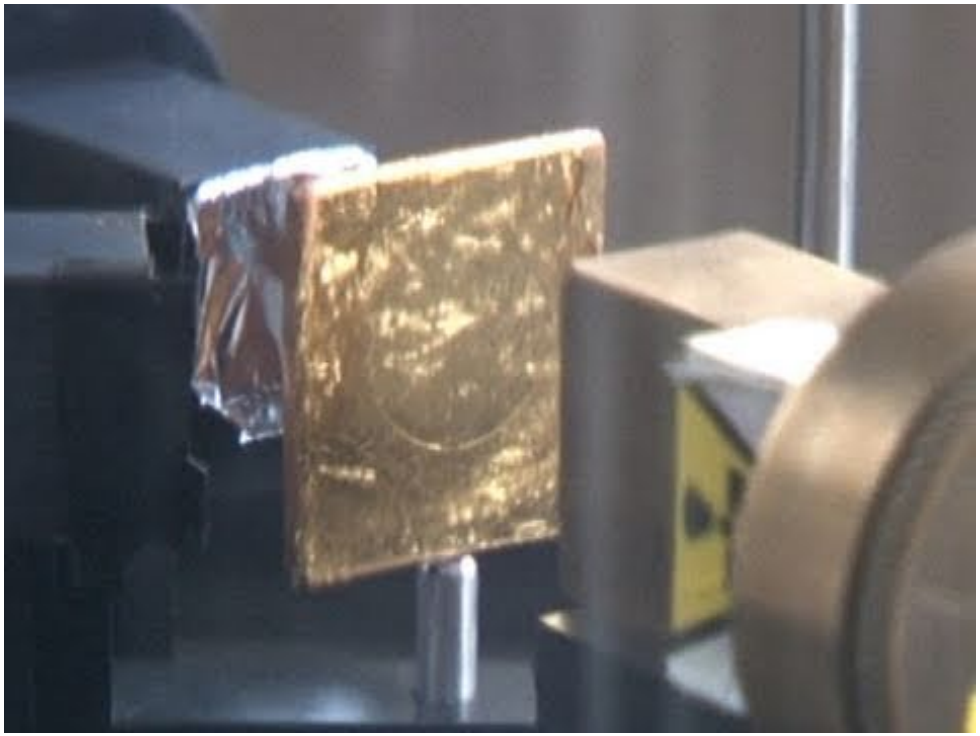
FIGURE 38.11 Rutherford's experiment to shoot high-speed alpha particles through a thin gold foil.



Large Angle Scattering



"It was quite the most incredible event that has ever happened to me in my life. It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you." - Ernest Rutherford



Backstage Science: Rutherford's Gold Foil Experiment

https://www.youtube.com/watch?v=XBqHkraf8iE

Credit: STFC

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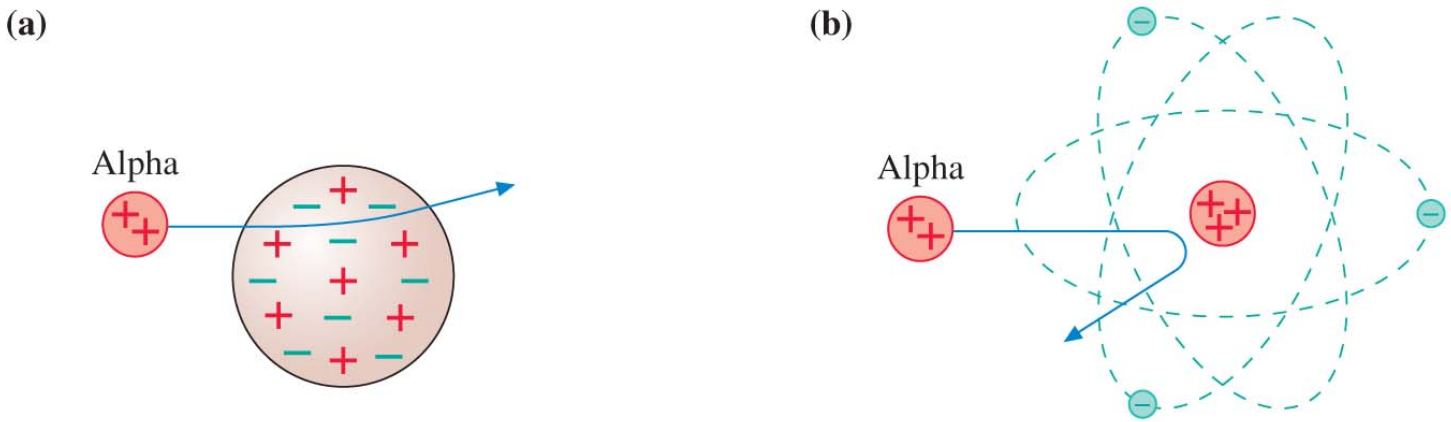
FIGURE 38.12 Alpha particles interact differently with a concentrated positive nucleus than they would with the spread-out charge in Thomson's model.



The alpha particle is only slightly deflected by a Thomson atom because forces from the spread-out positive and negative charges nearly cancel.

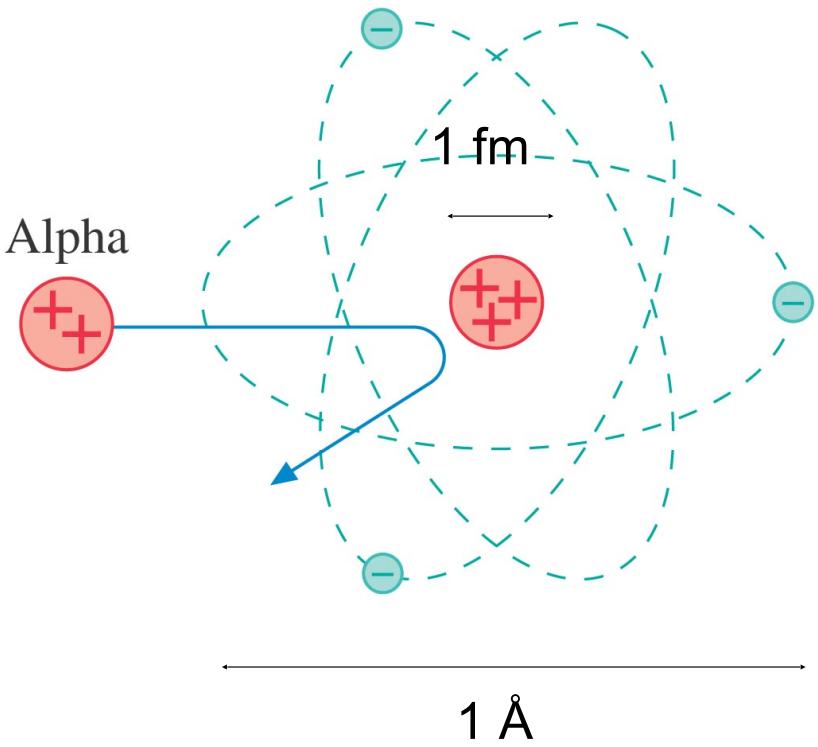
If the atom has a concentrated positive nucleus, some alpha particles will be able to come very close to the nucleus and thus feel a very strong repulsive force.

- In Plum Pudding model, the alpha particles should not be deflected.
- Deflection through large angles suggested that atoms contain a positive core.

Rutherford's nuclear model of the atom = "Planetary Model"

The atomic nucleus has a diameter of ~ 10^{-15} m =1 fm

The diameter of the atom is ~ 10^{-10} m = 1 Å



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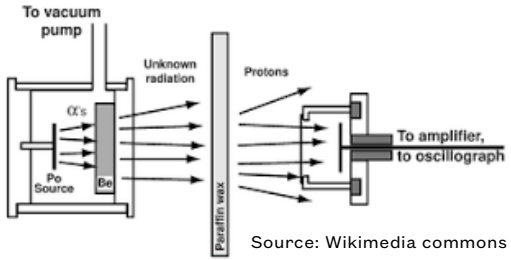
Protons and Neutrons



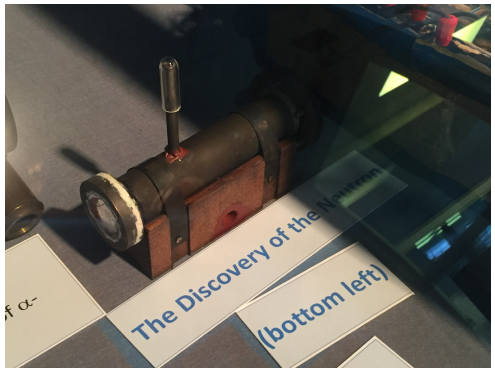
Source: Wikimedia commons

Rutherford's discovery of the nucleus, +ve charge was later recognized to be associated with a subatomic particle → proton

Nearly all the mass of an atom is associated with the nucleus → proton is much more massive than electron

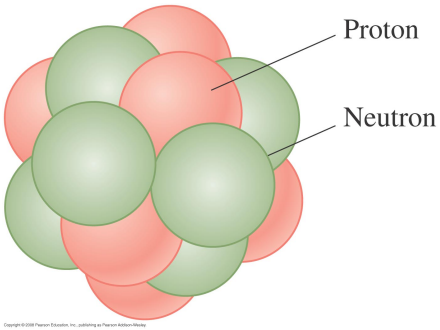
Determination of charge/mass ratio of various atoms suggested that the nucleus contained something else (in addition to protons).

The neutron was discovered in 1932 (by James Chadwick)



Chadwick's apparatus! (In Cavendish Museum)
Photo: S. Sheehy

Z= Atomic Number = # of protons
N = # of neutrons
A= Mass Number = Z + N



Chadwick, James (1932). "Existence of a Neutron". *Proceedings of the Royal Society A*. **136** (830): 692–708.

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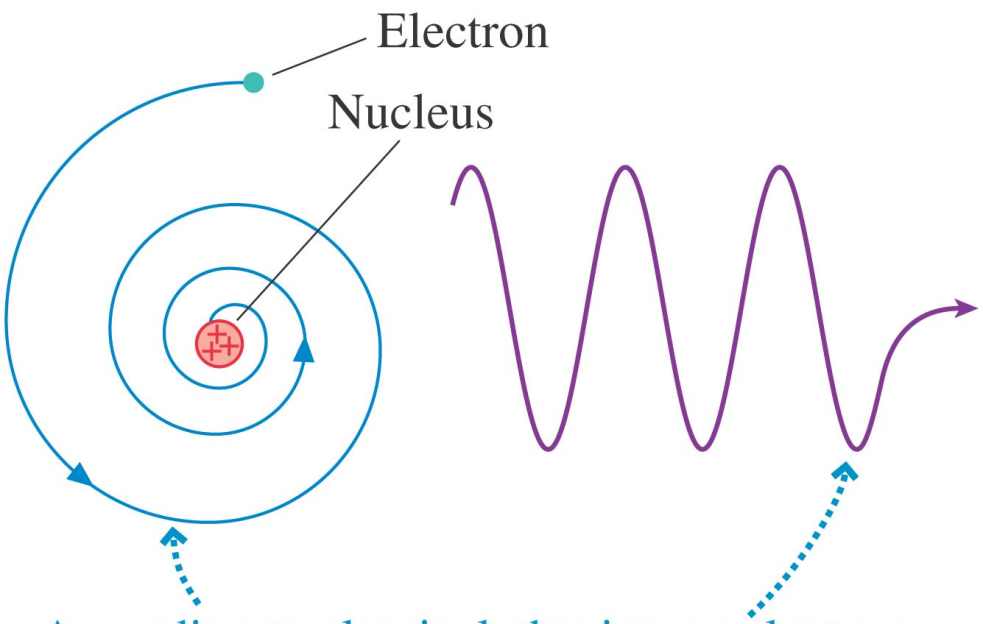
Classical Physics at the Limit

According to Maxwell's theory, the orbiting electrons in a Rutherford atom would emit EM radiation and lose energy… the solution to this problem lies with quantum physics.



According to classical physics, an electron would spiral into the nucleus while radiating energy as an electromagnetic wave.

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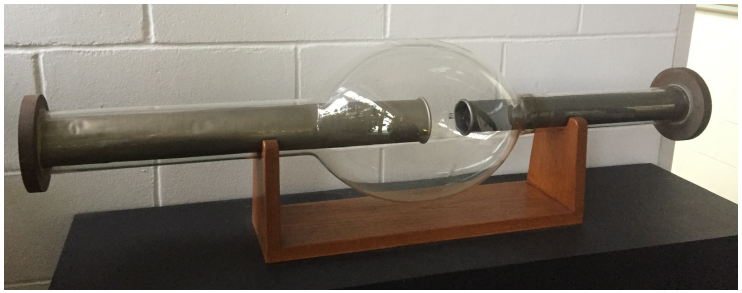
Key concept: The Electron Volt
useful unit of energy for atomic/nuclear systems

1 electron volt defined as the kinetic energy gained by an electron if it accelerates through a potential difference of **1 volt**.

$$E = q\,V$$

$$1 \text{ electron volt} = 1.60 \times 10^{-19} \text{ C} \times 1 \text{ volt}$$

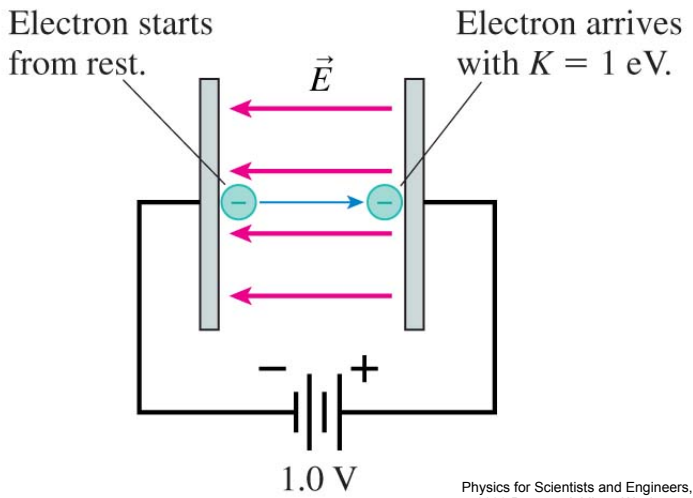
$$1\text{eV} = 1.60 \times 10^{-19} \text{ Joules}$$



The very first particle accelerator tube (at Cavendish Lab, Cambridge, by Cockcroft & Walton) where in the gap between metal tubes there was a voltage of up to 800kV

Photo: S. Sheehy

FIGURE 38.14 An electron accelerating across a 1 V potential difference gains 1 eV of kinetic energy.



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Example: energy of an electron

In a simple model of the hydrogen atom, the electron orbits the proton at 2.19×10^6 m/s in a circle with radius 5.29×10^{-11} m. What is the atom's energy in eV?

MODEL The electron has a kinetic energy of motion, and the electron + proton system has an electric potential energy.

SOLVE The potential energy is that of two point charges, with $q_{\text{proton}} = +e$ and $q_{\text{elec}} = -e$. Thus

$$E = K + U = \frac{1}{2}m_{\text{elec}}v^2 + \frac{1}{4\pi\epsilon_0}\frac{(e)(-e)}{r} = -2.17 \times 10^{-18}\ \text{J}$$

Conversion to eV gives

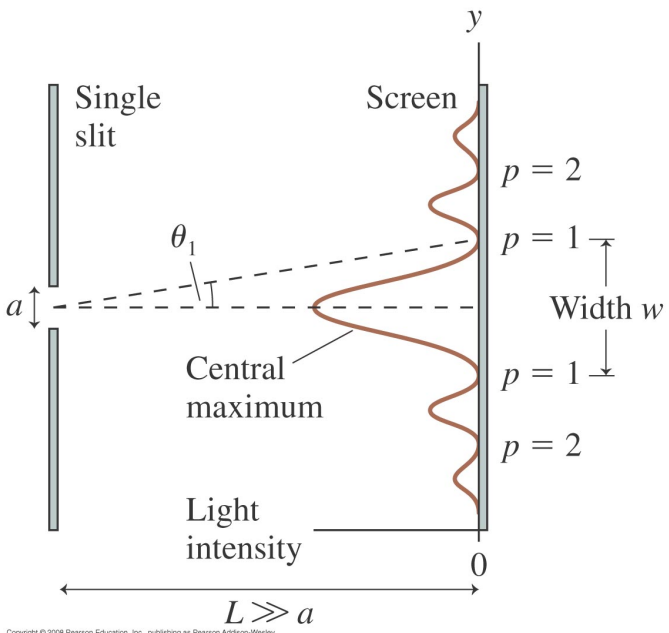
$$E = -2.17 \times 10^{-18}\ \text{J} \times \frac{1\ \text{eV}}{1.60 \times 10^{-19}\ \text{J}} = -13.6\ \text{eV}$$

ASSESS The negative energy reflects the fact that the electron is *bound* to the proton. You would need to *add* energy to remove the electron.

Wave-particle duality §38.3

Each photon produces a localized detection event on the screen (e.g. darkening of emulsion grain, click in photon counter). This is the <u>particle-like</u> behaviour.

The pattern on the screen is given by <u>wave</u> diffraction.



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Photons have both wave-like and particle-like properties.

Wave-particle duality §38.3

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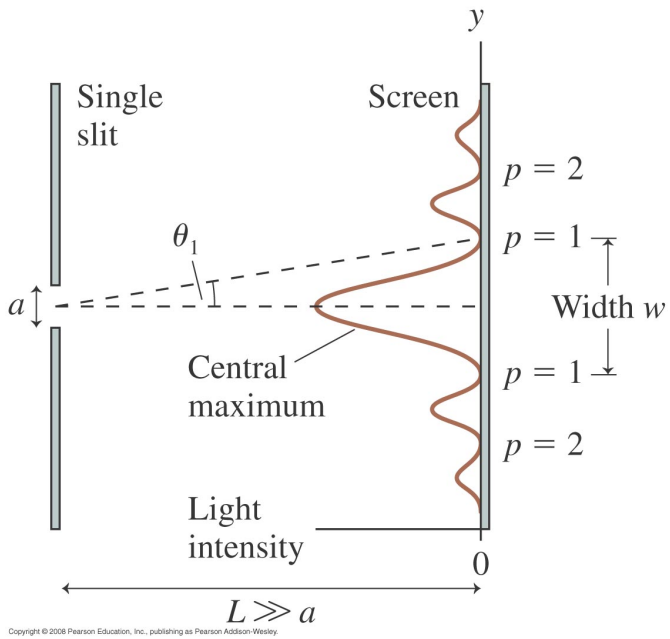

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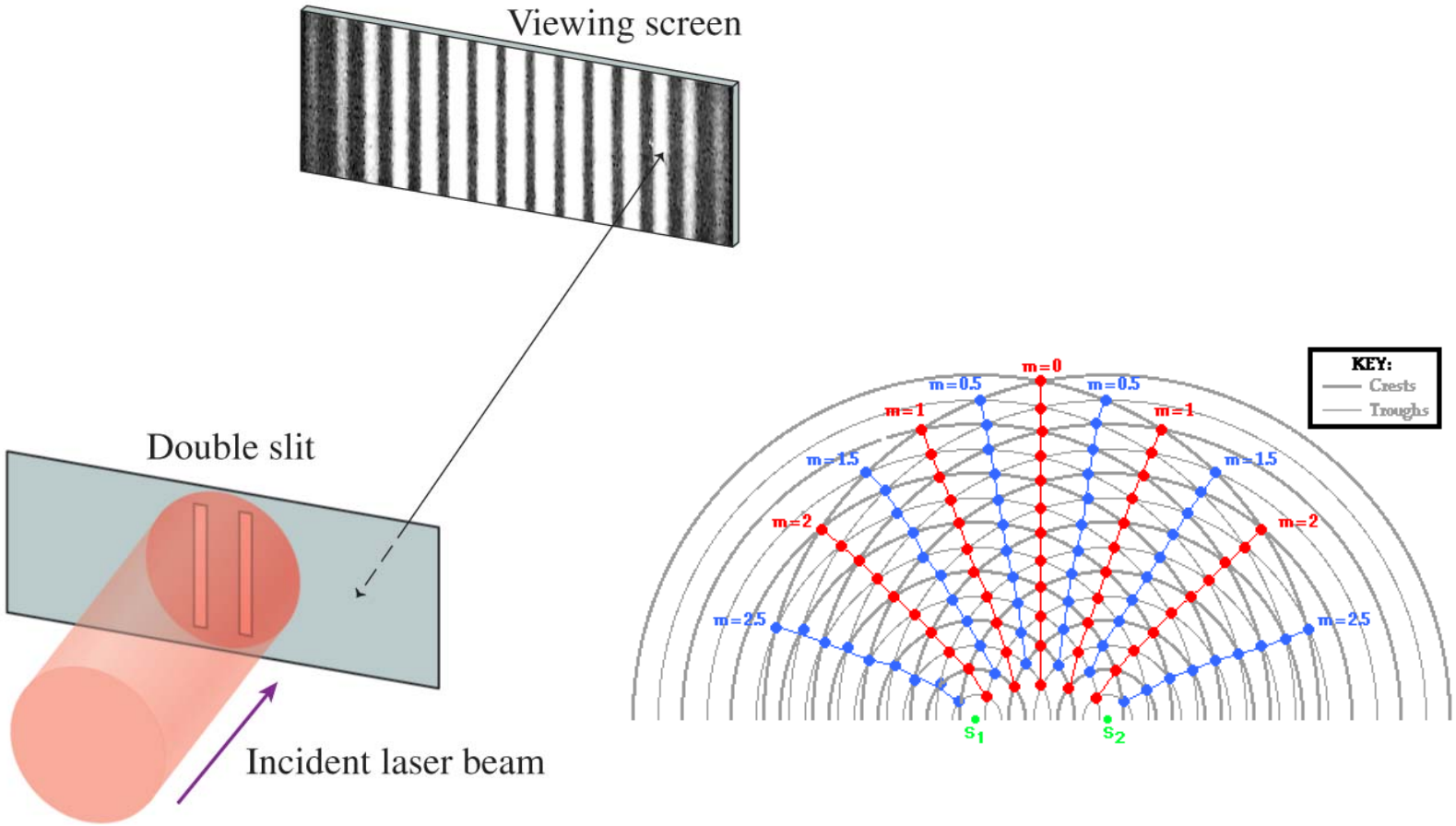
Photons have both wave-like and particle-like properties.

<u>Interpretation</u> – <u>Born</u>
The waves are waves of "probability" i.e. "intensity" means "probability"

Probability for a photon at point P ≈ $E^2(P)$.

FIGURE 38.2 Young's double-slit experiment showed that light is a wave.



Young's two-slit experiment:
→ Obtain interference pattern due to light waves from the left and right slits interfering constructively/destructively.

For interference to happen, the light wave must go through <u>both slits</u>.

The pattern is there also when the wave consists of a <u>single photon</u>, then we must conclude that the photon goes through <u>both</u> slits.

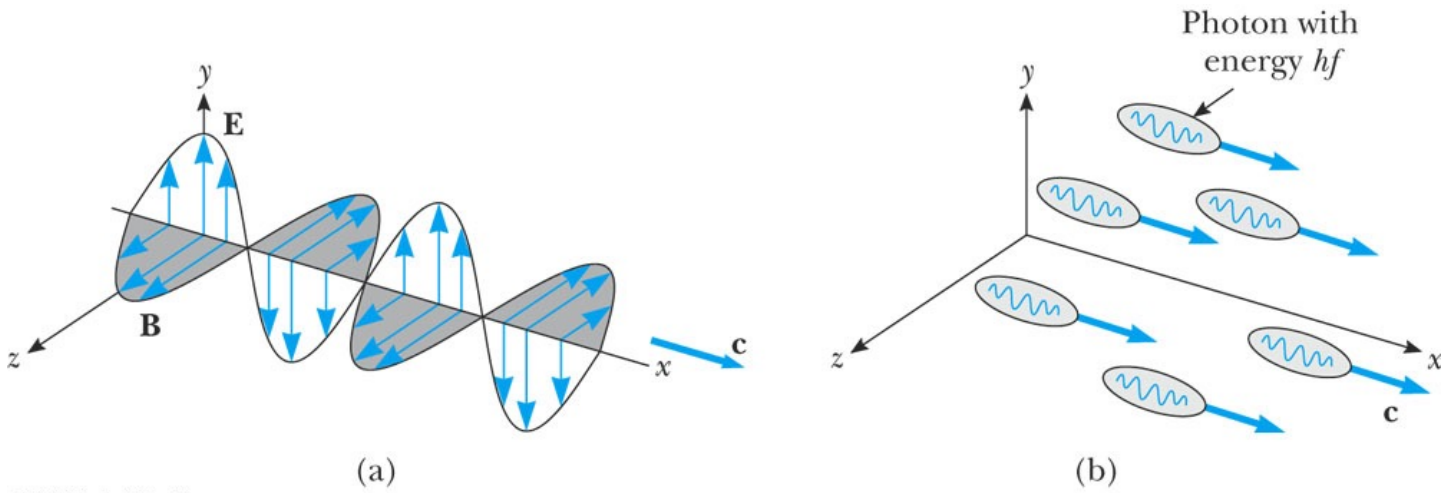
If you place a detector in front of slit 1 or 2, to observe which slit the photon actually goes through, you also destroy the interference pattern! (the photon is absorbed by the detector)

Photons are either waves or particles, but not both at once.

Photons

Classical EM wave

Photons: light quanta



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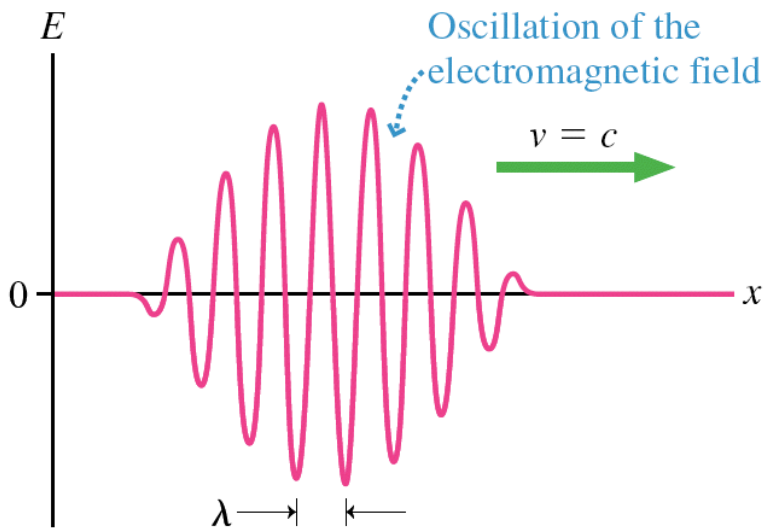
Photons

Photons are wave packets.

The electromagnetic wave shown has a wavelength and a frequency, yet it is also discrete and fairly localized.

FIGURE 39.11 A wave packet has wave-like and particle-like properties.



Wave-particle duality:

Wave to explain

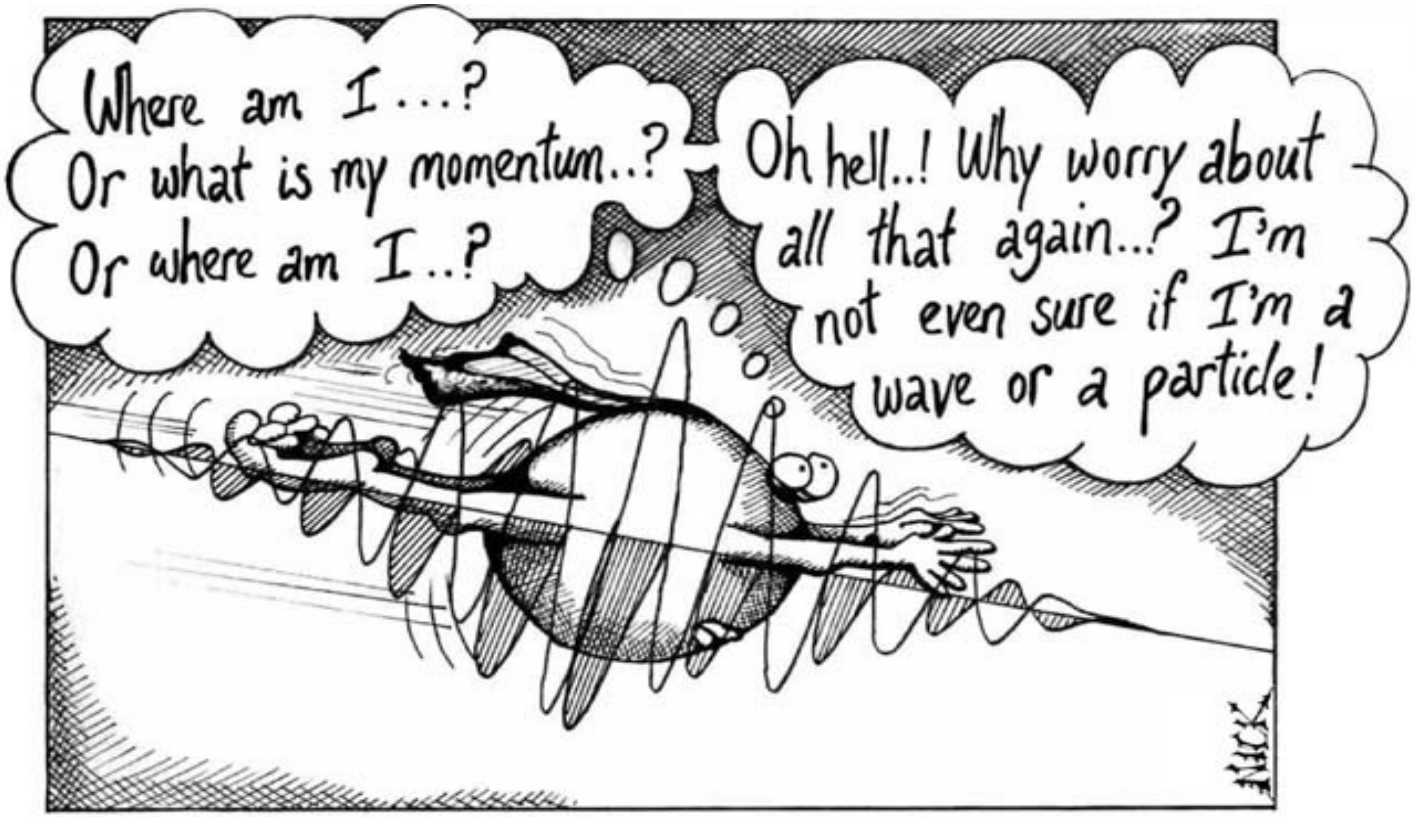
- refraction
- diffraction
- interference

Particle to explain

- photoelectric effect
- Compton effect

Photons give the detail

Waves provide the picture which accumulates over time



Photon self-identity problems.

Summary:

- Rutherford planetary model of the atom
- Classical physics limit
- Definition of electron volt
- Explain wave particle duality